Math Kangaroo Practice

Grade 3-4 · 5 questions · standard · seed 94380

Part A: Each question is worth 3 points

1. Sara made a tunnel out of identical cubes (fig. 1). Then Sara took it apart and built a complete pyramid (fig. 2) with some of the cubes. Nothing is hollow inside.

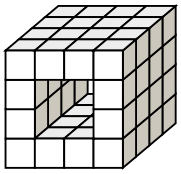


fig. 1

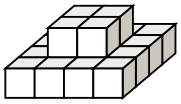


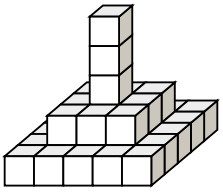
fig. 2

How many cubes from the tunnel were **not** used for the pyramid?

A. 16 ***B.*** 23 ***C.*** 24 ***D.*** 26 ***E.*** 28

Part B: Each question is worth 4 points

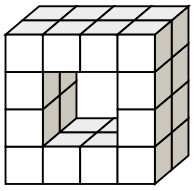
2. Nina built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 43 ***B.*** 42 ***C.*** 37 ***D.*** 35 ***E.*** 11

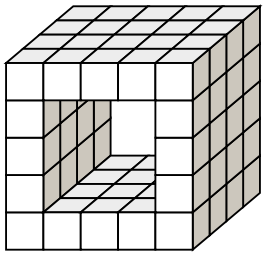
3. Tom built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Tom wants to fill the hole completely, so that the tunnel becomes one solid block.



How many more cubes are needed?

A. 4 ***B.*** 6 ***C.*** 8 ***D.*** 12 ***E.*** 24

4. Mia built this tunnel out of identical cubes. The hole goes straight through from the front to the back. Mia wants to fill the hole completely, so that the tunnel becomes one solid block.

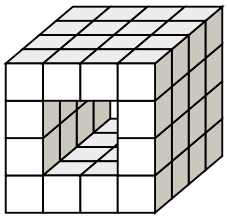


How many more cubes are needed?

A. 36 ***B.*** 28 ***C.*** 27 ***D.*** 12 ***E.*** 9

Part C: Each question is worth 5 points

5. Leo's tunnel is 4 cubes long. The hole goes straight through. Leo makes it longer by adding more cubes at the back in the same pattern, until it is 7 cubes long.



How many cubes does Leo have to add?

A. 12 ***B.*** 24 ***C.*** 34 ***D.*** 36 ***E.*** 84

Answer Key

1. **E**	**2. C**	**3. C**	**4. A**	**5. D**

Solutions

1. E (28) — Tunnel: 12 cubes in the front, 4 long: 12 × 4 = 48. Pyramid: 4 × 4 + 2 × 2 = 20. Left over: 48 – 20 = 28. [set-03/by_level/Grade_03-04/2002 Q20 · tunnel_to_pyramid]

2. C (37) — Count layer by layer, from the bottom: 5 × 5 + 3 × 3 + 1 × 1 × 3 = 37. [set-03/by_level/Grade_03-04/2002 Q20 · count_pyramid]

3. C (8) — The hole is 2 × 2 = 4 cubes in the front and goes 2 cubes back: 4 × 2 = 8. [set-03/by_level/Grade_03-04/2002 Q20 · fill_the_hole]

4. A (36) — The hole is 3 × 3 = 9 cubes in the front and goes 4 cubes back: 9 × 4 = 36. [set-03/by_level/Grade_03-04/2002 Q20 · fill_the_hole]

5. D (36) — Each slice of the tunnel has 4 × 4 – 2 × 2 = 12 cubes. 3 new slices: 12 × 3 = 36. [set-03/by_level/Grade_03-04/2002 Q20 · longer_tunnel]